

Wildfire Risk Mitigation in Bonner County, Idaho

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For Dr. Jenna Tilt
GEOG 551
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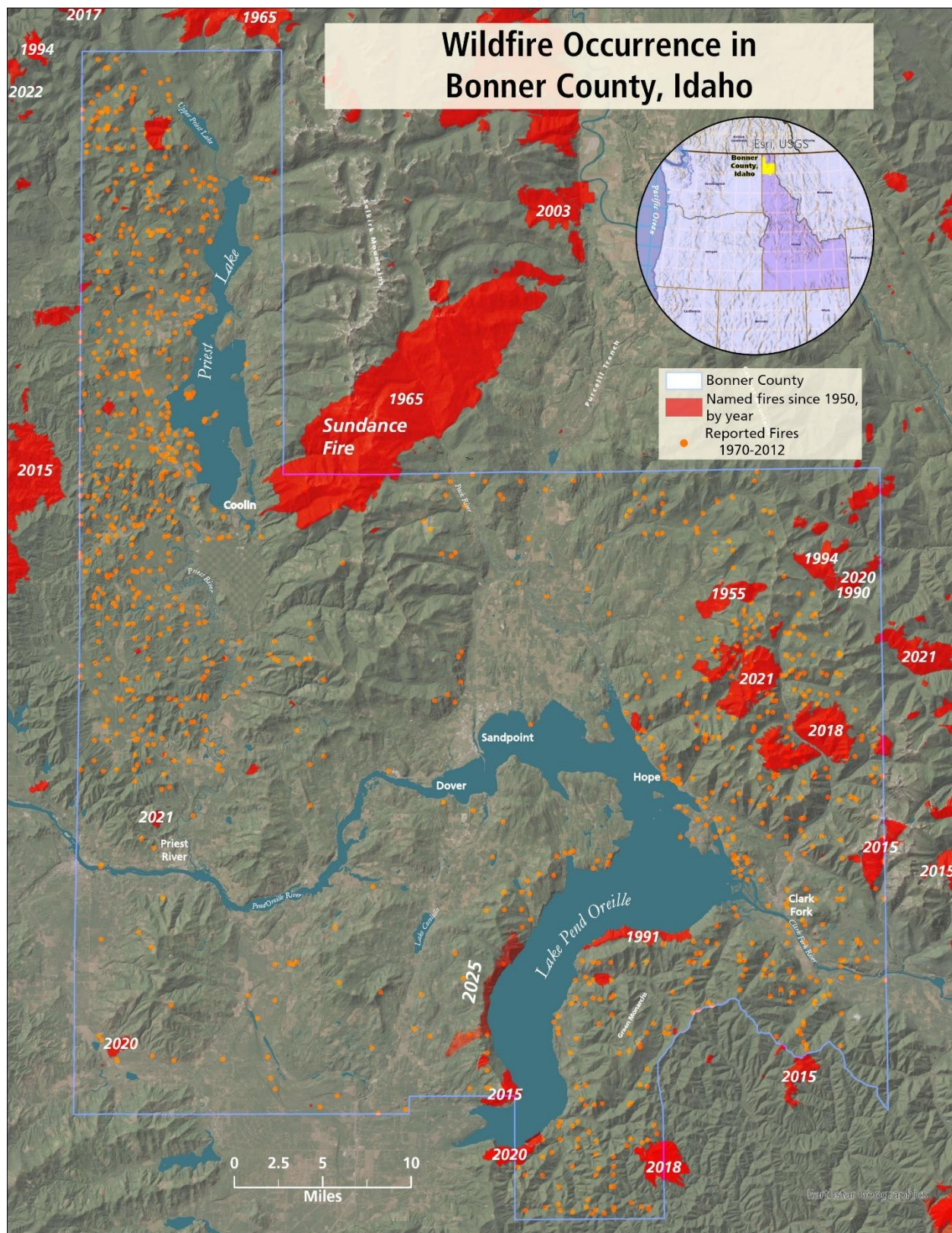


Figure 1, Wildfire Occurrence, NFIC, USFS, ArcGIS Pro

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Chapter 1. Baseline Conditions

1.1 Study area

Bonner County, Idaho had a total population of 47,110 in 2020 residing within its 1918.33 sq. miles, for a population density of 27.2 people per square mile (US Census Bureau, 2021). Bonner County has two of the three biggest lakes in Idaho, Lake Pend Oreille and Priest Lake, as well as Idaho's largest ski resort, Schweitzer Mountain. Elevations range from 2053 feet at Oldtown, to over 7000 at Scotchman's Peak. About 40% of Bonner County is private land. 44.4% of the land is federally owned, the state of Idaho owns another 15.2%, while municipalities only account for 1% of the total area. There are thirteen named communities with permanent residents, while only about 15% of the county is zoned for residential use (urban or rural) (Comp Plan Update, 2024). All told, more than 70% of the county is beneath forest canopy (Data | Multi-Resolution Land Characteristics (MRLC) Consortium, 2024).

Between 1970 and 2012, 1409 fire starts were recorded in Bonner County by the US Forest Service, an average of 33 per year (Idaho Panhandle National Forests - Geospatial Data, 2025). Two of the most studied fires of the 20th century burned at least partially in Bonner County: The Great Fire of 1910 that burned nearly 3 million acres in Idaho and Montana (The Great Fire of 1910, n.d.), and the Sundance fire of 1967, which at the time set the standard for extreme fire behavior. "In the nine-hour period from 1400 to 2300, the fire increased by 50,000 acres and advanced 16 miles to the northeast, with long-range spotting up to 10 miles" (Sundance Fire (Idaho) - September 1, 1967, 2024). See figure 1.

In this report, wildfire *hazard* refers to the biophysical conditions that enable fire occurrence and spread, including climate, fuels, and topography. *Risk* reflects the likelihood that wildfire will impact human assets, particularly residential structures, given those hazards and patterns of exposure. *Vulnerability* describes the social and structural characteristics that shape

how households experience and recover from wildfire impacts. The following analyses distinguish these concepts to clarify where wildfire is most likely, where it most threatens homes, and which populations are most affected.

This chapter describes the wildfire hazard in Bonner County, both in terms of historical data and with regard to projected changes in fire behavior. The fire return interval in the Rocky Mountain Montane Province has been altered by human interventions. Recent changes in the temperature and precipitation regime both locally and nationally suggest fires on average are becoming more destructive and faster moving.

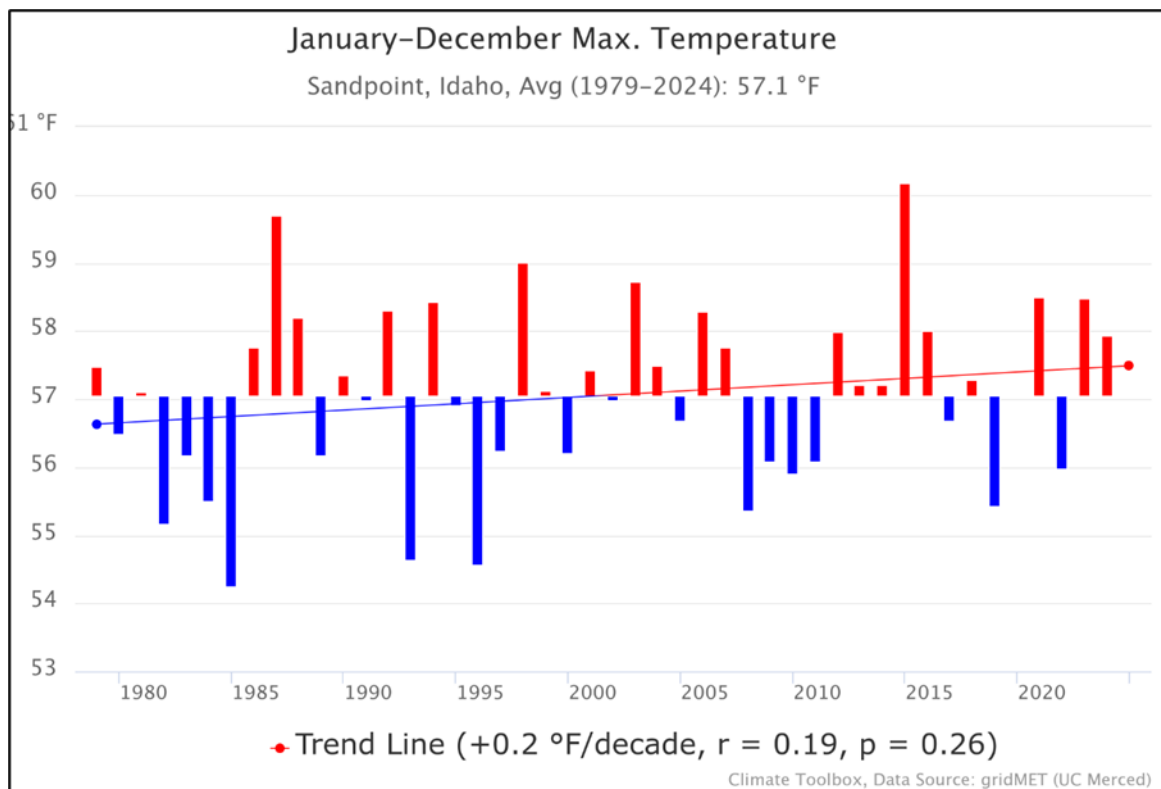


Figure 2, Average annual maximum temperatures for Sandpoint, Idaho (1979 -2024).

Together, the historical fire records establish Bonner County as a landscape shaped by high-frequency ignition and potential for extreme fire behavior (Dalton et al., 2013). While total annual precipitation is not predicted to increase or decline markedly, annual average temperatures in the county have gone up at least 1°C since 1979 (Hegewisch & Abatzoglou,

n.d.). Nationwide, fire behavior has changed in response to climate and human inputs (Balch et al., 2024), with unprecedented rates of spread. The Sundance Fire was an early example of fire behavior that has become commonplace in this century. Whether Bonner County's population will continue to grow in the coming years remains uncertain, but current development patterns place more people and structures within fire-prone landscapes. This analysis is therefore framed around "fast fires," defined here as wildfire events that grow more than 4,000 acres in a single day.

Parcel-level wildfire risk was modeled using national interagency fire center raster datasets that characterize burn probability, fire intensity, and structure-relevant exposure. These raster inputs were combined and normalized to derive two composite indicators used throughout the analysis (Dillon et al., 2024). Parcel Risk to Homes (PRH) represents the relative likelihood that wildfire will affect residential structures on a given parcel, while Treatment Priority near Homes (TPN) identifies parcels where landscape-scale fuel treatments are most likely to reduce risk to nearby homes (Dillon et al., 2024). Evaluating these metrics allows wildfire hazard to be evaluated in relation to development patterns, producing a spatially explicit picture of risk across Bonner County.

Using comparable national methods, recent estimates suggest that approximately 98% of Bonner County residences are exposed to some level of wildfire risk (Fire Factor® Wildfire Risk Model Methodology, 2023). Local population growth over the past decade further increases the potential consequences of wildfire events (Comp Plan Update, 2024), particularly given that human-caused ignitions tend to account for the majority of new wildfire starts (Balch et al., 2017).

Efforts are being made to implement fuels reduction strategies and defensible zones around homesites, but these measures are often insufficient as fire behavior gets more extreme. “... particularly given the rapid expansion of the WUI into areas with the greatest probability of wildfire (36, 37) and the mechanisms by which most homes burn. These fast fires represent only 2.7% of all events, yet they account for 89% of the total structures damaged or destroyed” (Balch et al., 2024).

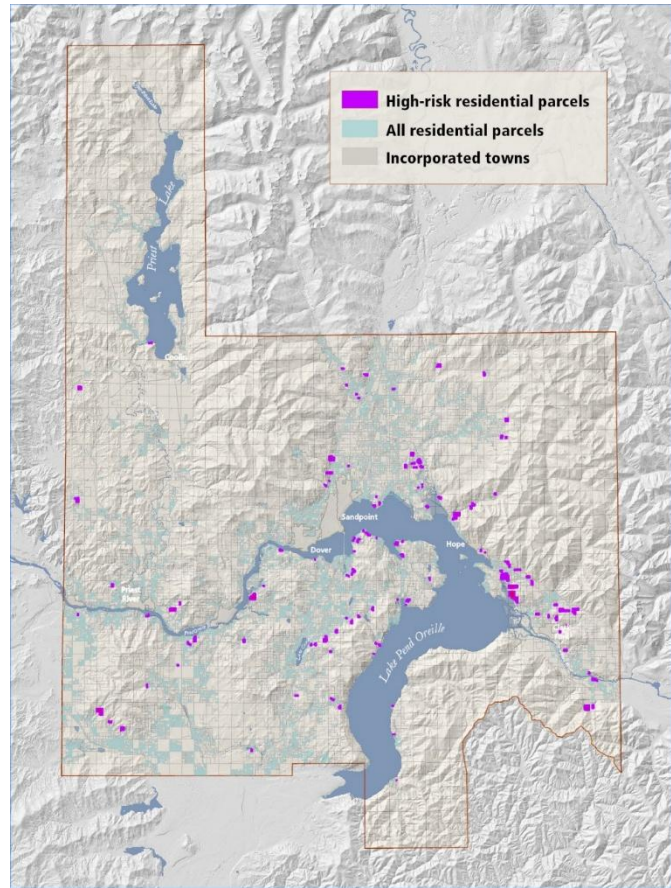


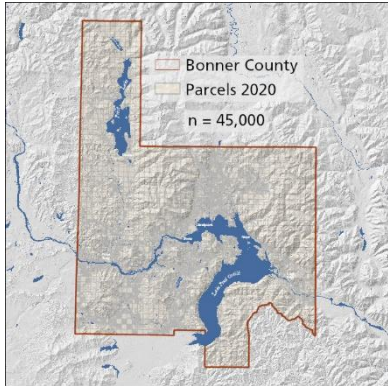
Figure 3, High-risk residential parcels in Bonner County, Idaho

1.2 Dataset Overview by Data Type

1. Vector Data - Parcels, Structures, and Administrative Features

Parcel Framework

- Bonner County parcel fabric (Bonner_2020; Mobile Map / Assessor data)
- Parcel geometry used as the primary spatial unit for analysis



Parcels and Land Use

County parcel fabric for Bonner County, Idaho

er_2020; Mobile Map / Assessor feed), used as the base unit for all parcel-level analyses. Parcel geometry defines analytic framework for integrating wildfire risk, demographic data, and emergency access across the county.

n = 45,000

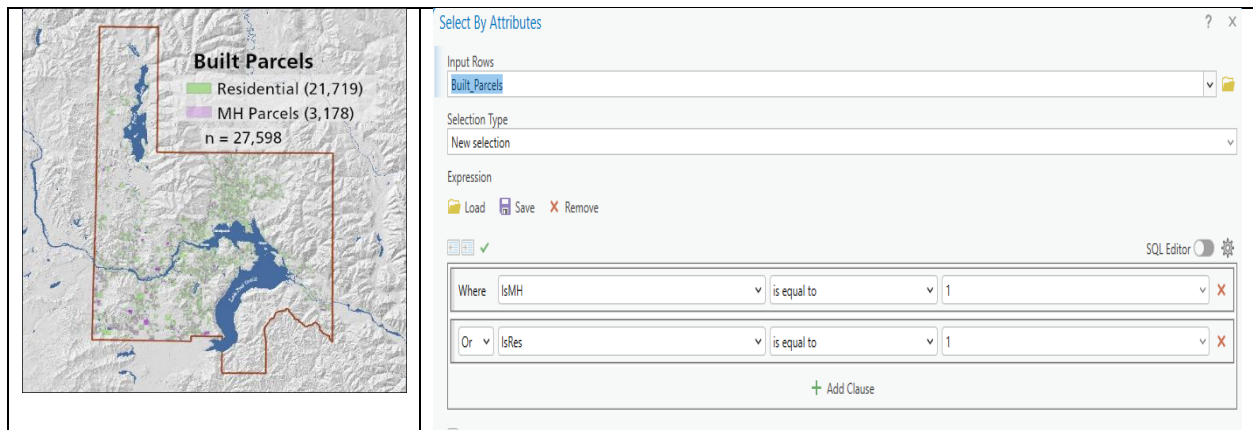
Built Environment Attributes

	• Parcels joined with Bing building footprint feature service • (n = 30,957 parcels with footprints)
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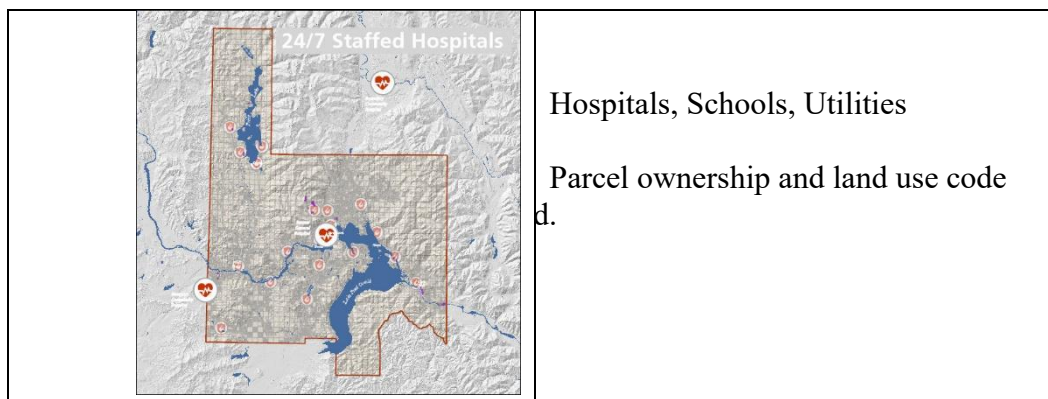
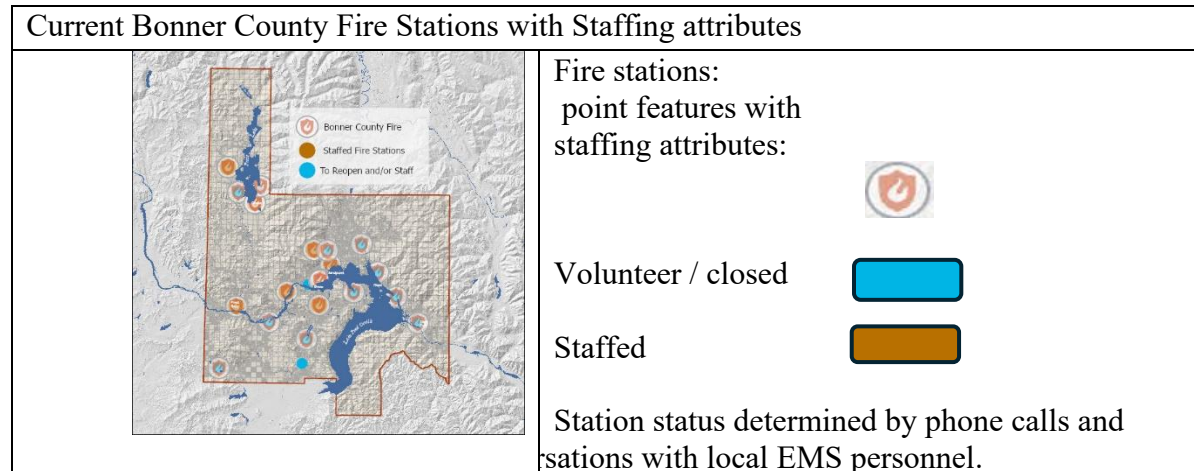
Residential classification derived from assessor property class descriptions:

IsRes - site-built residential 21,719

IsMH - manufactured housing 3,178



Critical Infrastructure



2. Demographic & Socioeconomic Data (Block-Level)

Primary Sources

- U.S. Census Bureau ACS 5-year estimates
- Esri Business Analyst demographic datasets (Enrich tool)

Population Sensitivity Metrics

- SNAP households
- Households with no health insurance
- Language isolation (Spanish / no English)
- Households with no vehicle: 869 HH in 304 blocks.
- Households with disability (county; ACS): 6,081 HH.
- Older adult living alone (65+ alone; county): 3,178 HH.
- Low-income households (WUI blocks): 1,842 HH in 137 blocks.

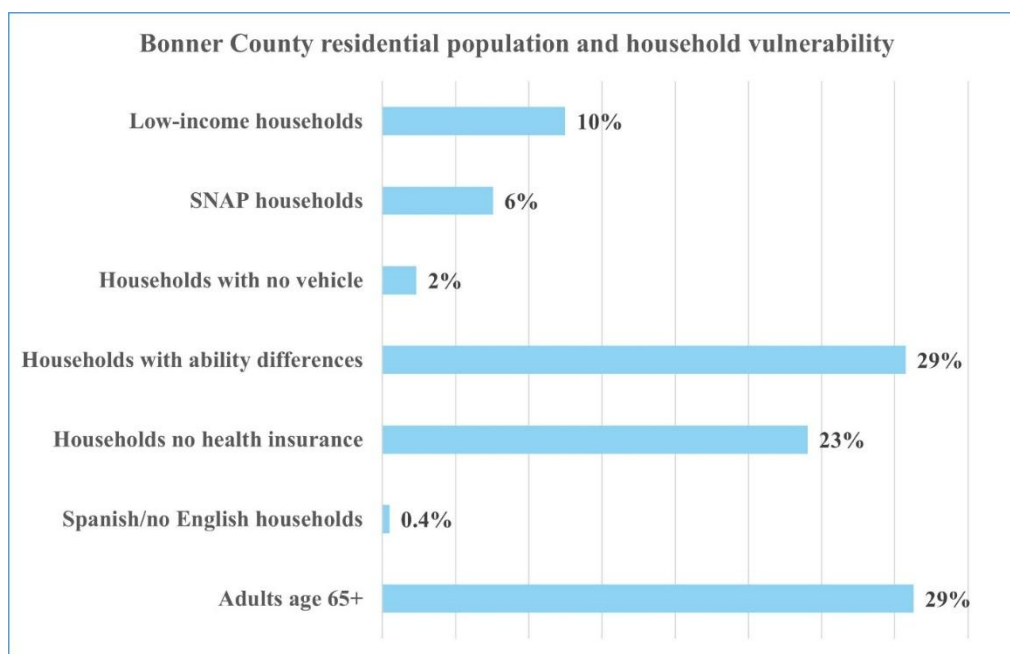


Figure 4, Residential population and household vulnerability

Note: Age, lack of insurance, and ability differences dominate vulnerability in wildfire-exposed residential areas; economic and linguistic vulnerability are present but smaller.

Table 1. Manufactured housing exposure within wildfire-exposed residential areas of Bonner County, Idaho.

Metric	Count	Unit
Residential parcels (total)	21,719	Parcels
Residential parcels in top-risk decile (PRH $\geq$ 90th percentile)	150	Parcels
Manufactured housing (MH) parcels (total)	3,178	Parcels
MH parcels in top-risk decile	14	Parcels

Share of high-risk parcels that are MH	9.3%	Percent
Share of MH parcels that are high risk	0.4%	Percent

Table 1, High-risk parcels by structure type

Table 2. Baseline social vulnerability and sensitivity metrics within wildfire-exposed residential areas of Bonner County, Idaho.

Sensitivity Metric	Count	Unit
Low-income households	2,153	Households
Snap households	1300	Households
Households with no vehicle available	399	Households
Households with ability differences	6,158	Households
Households with no health insurance	5,006	Households
Spanish/no English households	82	Households
Adults age 65+	14,920	Persons

Table 2, Baseline demographic sensitivity metrics.

Table 3. Vulnerability within high-risk residential parcels (calculated)

Sensitivity Metric	Count	Denominator	Percent
Household income below poverty level	243	3,040 HH	8.0%
SNAP Households	141	3,040 HH	4.6%
Households with no vehicle available	50	3,040 HH	1.6%
Households with ability differences	845	3,040 HH	27.8%
Households no health insurance	754	3,040 HH	24.8%
Spanish / no English Households	6	3,040 HH	0.2%
Adults age 65+	2,237	6,871 persons	32.6%

Table 3, High-risk parcel demographic vulnerability

Note: High-risk parcels are defined as residential parcels in the top decile (≥ 90 th percentile) of PRH. Household-level metrics use aggregated ACS 5-year estimates for census blocks intersecting high-risk parcels ($n = 3,040$ households); age-based vulnerability uses population counts ($n = 6,871$ persons).

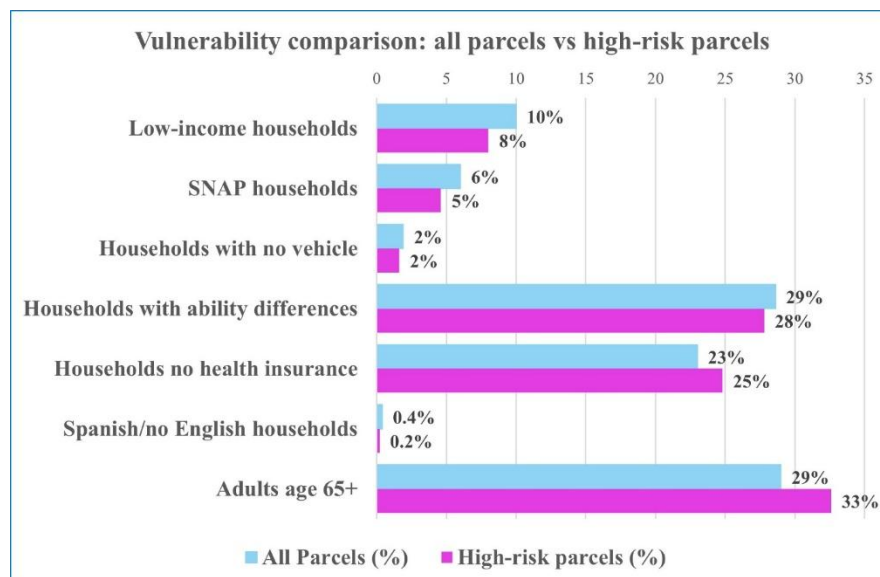
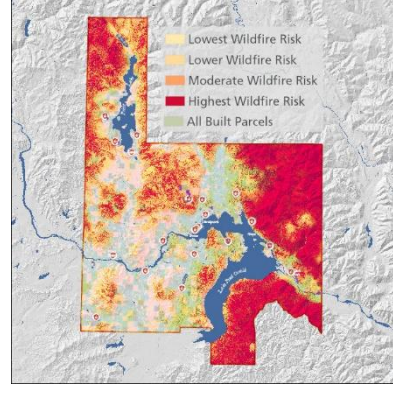


Figure 5 Parcel-level population vulnerability comparison

Note: Block-level population and household attributes were spatially allocated to high-risk residential parcels using areal apportionment, assuming uniform distribution of attributes within Census blocks.

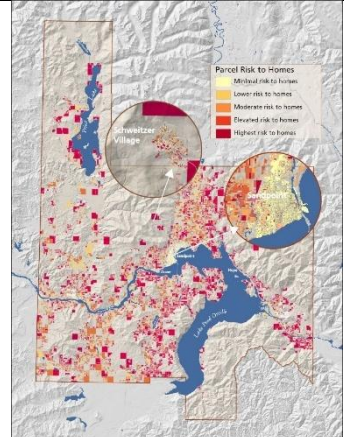
3. Raster Data: Wildfire Hazard, Probability, and Exposure

USDA Forest Service - Wildfire Risk to Communities [Wildfire Risk to Communities](#)

	Rasters used: WHP_ID: Wildfire Hazard Potential Relative likelihood of extreme fire behavior BP_ID: Burn Probability Long-term climatological likelihood of wildfire occurrence RPS_ID: Risk to Potential Structures Structure-adjacent fuels and exposure signal cRPS_ID :Contextualized / normalized RPS Percentile-based normalization for inter-parcel comparison CRPS90: 90th percentile contextual risk threshold Exposure_ID
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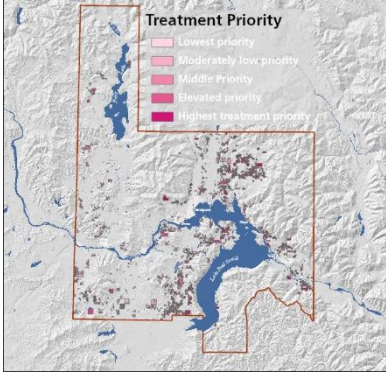
4. Derived Composite Metrics for treatment plans

Parcel Risk to Homes (PRH)

	A normalized, parcel-level index derived from USDA Forest Service <i>Wildfire Risk to Communities</i> structure-relevant risk rasters. Raster risk values were summarized to individual parcels and normalized to enable relative comparison and percentile-based targeting of residential wildfire risk. $PRH = ((0.7 * !Parcels_RiskNorm.RPS_Norm!) + (0.3 * !s_RiskNorm.cRPS90_N!)) * !Parcels_RiskNorm.exp_mult!$
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Treatment Priority near Homes (TPN)

Composite index identifying locations where landscape treatments most reduce risk to people and assets

	WHP_ID: Wildfire Hazard Potential Relative likelihood of extreme fire behavior across the landscape. BP_ID : Burn Probability Long-term climatological likelihood of wildfire occurrence. WHP × BP: Combined Hazard -Probability Surface Multiplicative combination of wildfire hazard potential and burn probability, representing locations where high-intensity fire is both likely to occur and likely to spread. $TPN = (0.6 * !whp90_n! + 0.4 * !bpmean_n!) * !exp_mult!$
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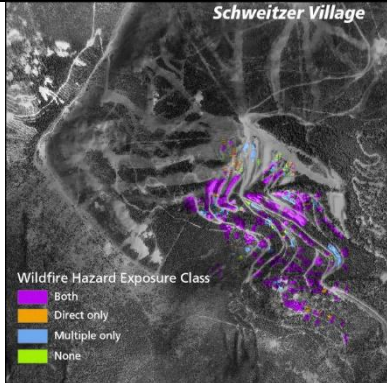
Derived Exposure & Classification Fields (Parcel-Level)

Exposure Indicators

- exp_dir_filled2 - Direct exposure (binary)
- exp_mult - Multiple / indirect exposure

Exposure Classes

- Both direct & multiple exposure 18,393 Parcels
- Direct only 543 Parcels
- Multiple only 4,806 Parcels
- None 1,153 Parcels

	exp_dir_filled2 - Direct Exposure Indicator parcel-level indicator identifying parcels with direct adjacency or proximity ential structures. exp_mult - Multiple / Indirect Exposure Indicator level indicator identifying parcels with multiple nearby residential es. Exposure_Class 18,393 Both 545 Direct only 4,806 Multiple 1,153 None
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Analytic Use

- Defines built and exposed parcel subsets
- Used to gate eligibility for PRH and TPN -based interventions

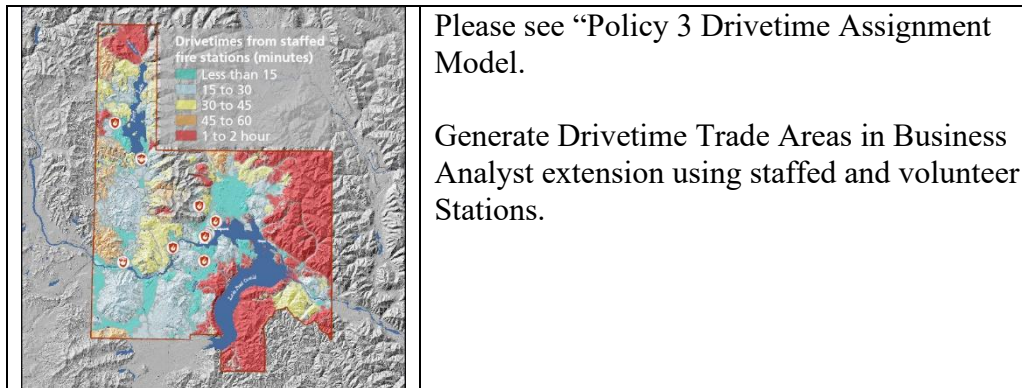
5. Network / Accessibility Data

Network Source

- Esri Business Analyst Network
- Data source: *United States (Esri 2025)*
- Network source: *Default*

Modeled Accessibility Outputs

- Parcel-to-fire-station travel times
- Drivetime bands:
 - 0-15 minutes
 - 15-30 minutes
 - 30-60 minutes
 - 60-120 minutes



Access baseline (current staffing)

Populations within fire-station access bands for all 30,957 parcels with structures (at current staffing):

- 0 -15 min: 18,272
- > 15 min: 6,248

Chapter 2. Final Goals and Methods

2.1 Goals

This project’s goals are to:

1. Reduce wildfire risk to homes at minimized cost by directing fuels or home hardening treatments to where they most reduce, while prioritizing areas with the highest treatment leverage near homes (TPN). PRH and TPN are designed as relative planning indices
2. Improve equitable emergency access through staffing and operational changes that place more residents within a 0–15-minute response window, with particular attention to mobile homes and low-income WUI blocks.

2.2 Policy Objective 1 - Targeted fuels work near homes (TPN hotspots)

Policy 1 asks “Where are fuels treatments needed?” and focuses county fuel-treatment efforts on locations where landscape work provides the greatest protection to people and property. Target parcels are defined as the top quintile of Treatment Priority near Homes (TPN) values (~4,554 parcels), representing areas with both high wildfire potential and direct or multiple exposure to nearby residences. Importantly, this treatment set includes ~608 manufactured-home parcels, indicating meaningful benefits for more vulnerable housing types. In practical terms, Policy 1 concentrates fuel-reduction dollars where they most measurably reduce structure exposure (Barker et al., 2025).

2.2a

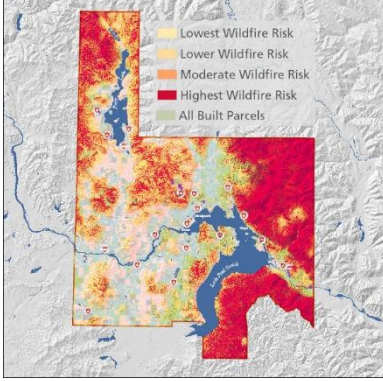
TPN hotspot parcels are identified by selecting parcels with TPN values at or above the fifth-quintile threshold and requiring modeled exposure to residential structures. Risk reduction is simulated by applying a calibrated reduction factor to baseline PRH values for treated parcels, while non-treated parcels retain baseline risk.

Policy 1 Methods

Data used for TPN:

Raster inputs were obtained from the U.S. Forest Service Wildfire Risk to Communities (WRTC) dataset (version/year), including WHP_ID (wildfire hazard potential), BP_ID (burn probability), and Exposure_ID (structure exposure/adjacency). Parcel geometries were obtained from Bonner County built-parcel data.

Raster Data:

	Rasters used: WHP_ID: Wildfire Hazard Potential Relative likelihood of extreme fire behavior BP_ID: Burn Probability Long-term climatological likelihood of wildfire occurrence Exposure_ID: structure exposure/adjacency USDA Forest Service - Wildfire Risk to Communities Wildfire Risk to Communities
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Vector Data: Parcel_RiskNorm_BASE

Summary of Transformations

Transformation Type	Purpose	Tool
Raster normalization	Standard hazard inputs	Raster Calculator
Raster weighted overlay	Create TPN index	Raster Calculator
Raster → vector extraction	Move hazard to parcel scale	Extract Multi Values to Points
Reclassification (quintiles)	Identify priority treatment areas	

Table 4, TPN Data Transformations

Calculate Field	$TPN = (0.6 * !whp90_n! + 0.4 * !bpmean_n!) * !exp_mult!$
-----------------	--

TPN Weighting strategy:

Component	Weight	Rationale
Wildfire Hazard Potential (whp90_n)	0.6	Represents fire intensity potential and flame length drivers
Burn Probability (bpmean_n)	0.4	Represents likelihood of fire occurrence

Table 5, TPN Weighting Strategy

Wildfire Hazard Potential (WHP) was weighted more heavily (0.6) than Burn Probability (BP) (0.4) because WHP integrates modeled fire behavior characteristics, including potential flame length and intensity, which more directly influence structural damage potential. While BP represents the likelihood of fire occurrence, WHP better captures the severity dimension of

hazard once ignition occurs. The weighting scheme therefore emphasizes fire behavior intensity over ignition frequency while still retaining BP as a meaningful component of overall hazard. This balance reflects the objective of prioritizing treatment in areas where fire intensity near structures poses the greatest potential consequence.

TPN Workflow (ArcGIS Pro)

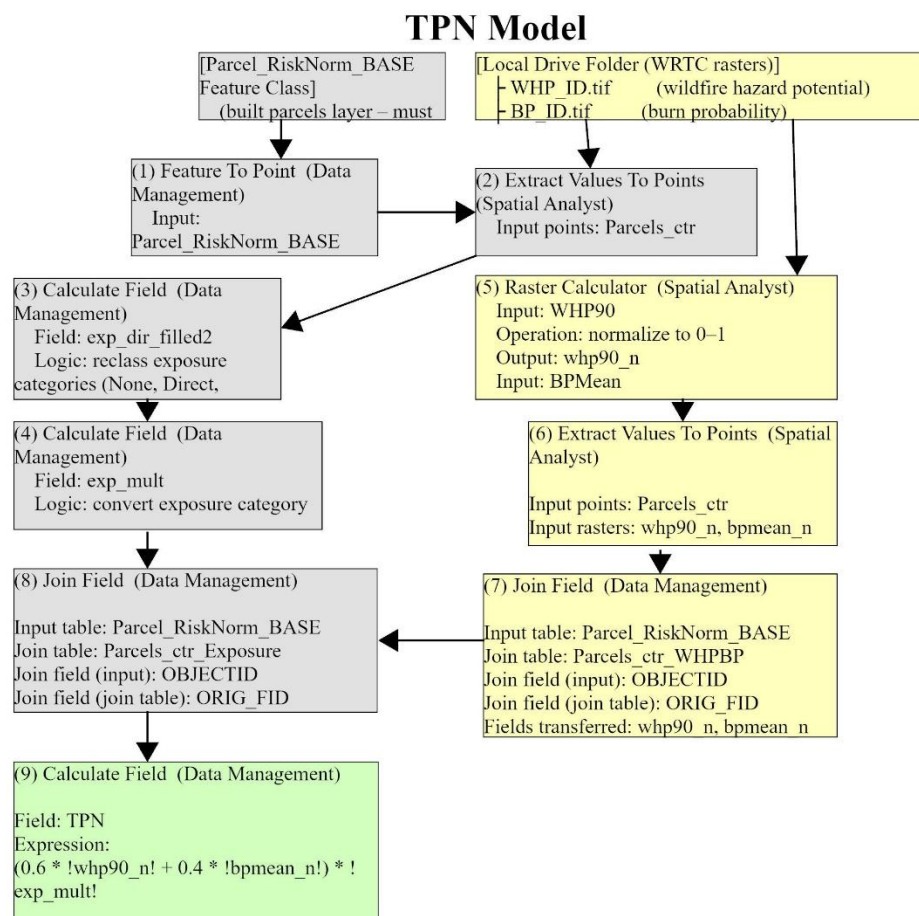


Figure 6, TPN workflow diagram

Model Limitations and Assumptions

This model assumes that raster-based hazard values adequately represent parcel-scale conditions, although intra-parcel heterogeneity may not be fully captured through centroid extraction. The

exposure multiplier simplifies adjacency dynamics into categorical classes, which may not fully reflect complex structure-to-structure ignition pathways. Hazard surfaces are treated as static and do not account for temporal variability in fuel moisture, suppression response, or extreme weather events. The normalization process standardizes values for comparability but may compress meaningful variation at the upper and lower extremes of the distribution. Finally, the weighting scheme reflects planning judgment rather than empirical calibration, and alternative weight combinations could produce different prioritization outcomes.

2.3 Policy Objective 2 - Home-hardening at the highest-risk homes (PRH hotspots)

Policy 2 targets parcel-scale mitigation where structure-specific wildfire risk is highest. The treatment cohort consists of residential parcels in the top quintile of PRH values (n = 970 treated parcels). Modeled upgrades include ember-resistant vents, Class-A roofing, non-combustible 0 -5 ft zones, and targeted access improvements (Barrett, 2024).

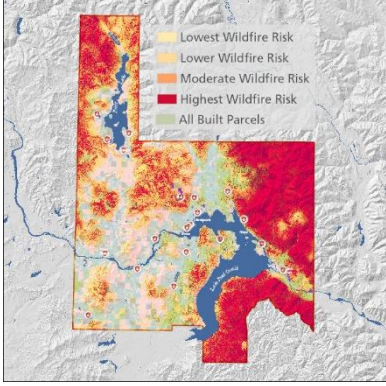
Across treated parcels, the modeled effect is a mean 20% reduction in parcel hazard (n = 970). While minor variation exists (three parcels at 10 -15%), 99.7% of treated parcels show a 20% modeled reduction. This reduction reflects decreased structure ignitability rather than landscape fuel modification. Because the treatment effect is applied uniformly within the targeted cohort, the policy produces consistent hazard reduction across high-risk homes.

Manufactured-home parcels are represented within the treated group, making this policy particularly relevant in areas where landscape-scale fuel treatments are constrained and structure vulnerability dominates risk.

2.3a

Data used for PRH

Raster Data:

	Rasters used: RPS_ID: Risk to Potential Structures integrates burn probability and fire behavior to estimate potential structural impact. cRPS90: 90th percentile contextual structural risk threshold) Exposure_ID: structure exposure/adjacency USDA Forest Service - Wildfire Risk to Communities Wildfire Risk to Communities
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Vector Data: Parcel_RiskNorm_BASE

Summary of Transformations

Transformation Type	Purpose	Tool
Raster Normalization	Standardize RPS surface to 0–1 scale	Raster Calculator
Contextual thresholding	Capture high-end structural risk clusters	Raster Calculator
Raster > vector	Move normalized risk values to parcel centroids	Extract Values to Points
Attribute Join	Attach structural risk metrics to parcels	Join Field
Weighed overlay	Blend structural and contextual risk	Calculate Field
Exposure Adjustment	ncorporate structure adjacency effects	Calculate Field
Reclassification (quintiles)	Identify PRH hotspot cohort	Symbology Reclassify

Table 6, PRH Data Transformations

Calculate PRH	$PRH = ((0.7 * !RPS_Norm!) + (0.3 * !cRPS90_N!)) * !exp_mult!$
---------------	---

PRH Weighting Strategy

Component	Weight	Rationale
Structural Risk (RPS_Norm)	0.7	Captures parcel-specific wildfire impact potential derived from WRTC

Contextual Structural Risk (cRPS90_N)	0.3	Emphasizes high-end structural risk clusters and spatial concentration of hazard
Exposure Multiplier (exp_mult)	Multiplicative modifier	Adjusts structural risk based on adjacency and structure-to-structure exposure

Table 7, PRH Weighting Strategy

The PRH model assigns 70% weight to parcel-level structural risk (RPS_Norm) and 30% weight to contextualized high-end structural risk (cRPS90_N). The heavier weighting of RPS_Norm reflects the primacy of direct structure-specific wildfire impact potential derived from the Wildfire Risk to Communities dataset. The inclusion of the 90th-percentile contextual term captures spatial clustering of extreme structural risk, ensuring that parcels embedded within high-risk structural environments receive additional emphasis. The exposure multiplier (exp_mult) is applied multiplicatively to reflect adjacency-driven amplification of structural vulnerability rather than functioning as an additive risk component.

PRH Workflow (ArcGIS Pro)

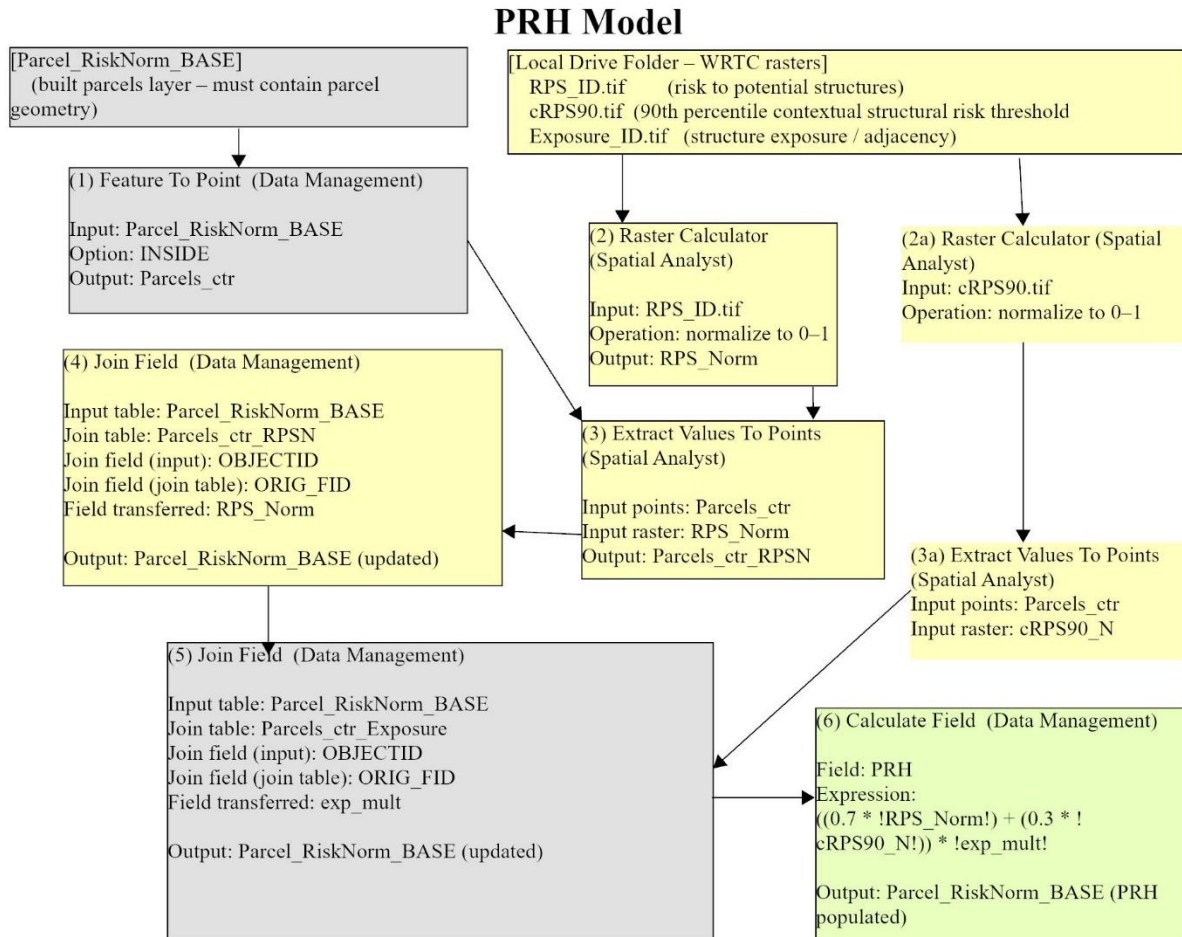


Figure 7, PRH Workflow diagram

Model Limitations and Assumptions

The PRH model assumes that the Wildfire Risk to Communities RPS surface accurately represents parcel-level structural wildfire impact potential at the spatial resolution used. The 0–1 normalization presumes comparability across parcels but does not alter underlying uncertainty in modeled fire behavior inputs. The contextual 90th-percentile term assumes that spatial clustering of high structural risk increases vulnerability, though the 30% weighting is a policy-informed modeling decision rather than an empirically calibrated coefficient. The exposure multiplier simplifies complex structure-to-structure and structure-to-fuels interactions into discrete categories, which may underrepresent fine-scale defensible space variation or building material

differences. Additionally, the model is static and does not incorporate dynamic suppression response, weather variability, or temporal changes in fuels or development patterns.

2.4 Policy 3 Staff and reopen stations to reduce emergency response times

Policy 3 evaluates the effect of staffing and reopening fire stations to reduce emergency response times across Bonner County. The policy simulates a scenario in which all existing stations are staffed to a minimum 24/7 engine standard, including recently closed or volunteer-only stations (e.g., Careywood). The goal is to shift parcels from long response times into faster service bands where initial attack and evacuation support are most effective (Rahn, 2010).

Under the all-stations-staffed scenario, the number of parcels within a 0 -15-minute response window increases from 18,272 to 24,520 - a gain of 6,248 parcels. Additionally, approximately 1,966 parcels, including ~206 mobile homes, move from response times of 30 - 120 minutes into ≤ 30 minutes, substantially improving emergency access for residents in remote corridors. These patterns align with recent wildfire experience in the county, including the Sunset Fire, where delayed arrival constrained suppression and evacuation support.

A small number of parcels (~96) remain classified as “no service” due to geographic isolation and network limitations. These locations are unlikely to benefit from staffing changes alone and instead require alternative strategies, such as interagency response agreements, water access improvements, or community-scale mitigation measures.

Implementation logic.

Policy 3 compares parcel-level emergency access under two scenarios: baseline staffing and an all-stations-staffed configuration. Changes in response time are summarized using both categorical drive-time bands and ranked improvement metrics to capture who benefits, how

much, and where equity-relevant gains occur. Staffing cost estimates are based on reported levy fact sheet figures for 24/7 staffing scenarios in Bonner County (Welch, 2025).

2.4a Policy 3 Methods

Data used for drivetimes

Vector Data;

- Fire stations feature class with staffing attribute
(0 = volunteer/closed, 1 = 24/7 staffed, 2 = 40-hour)
- Business Analyst network service dataset (ArcGIS Online routing service)
- Parcel centroids (derived from Parcel_RiskNorm_BASE)
- Parcel_RiskNorm_BASE (to store access results and summaries)

Transformation Type	Purpose	Tool
Scenario filtering	Separate baseline vs staffed-station configurations	Select Layer by Attribute
Drive-time polygon generation	Create travel-time bands around stations	Generate Drive Time Trade Areas (Business Analyst)
Spatial overlay	Assign parcels to drive-time band	Spatial Join
Scenario comparison	Calculate change in drive-time classification	Calculate Field
Equity summarization	Count MH parcels and corridor-level gains	Summary Statistics

Table 8, Policy 3 Data Transformations

Drive-Time Band Construction

Drive-time service areas were generated using the **Generate Drive Time Trade Areas** tool in ArcGIS Business Analyst. Travel times were specified as cumulative thresholds of 5, 10, 15, 30, and 120 minutes from each active fire station. The tool produces overlapping polygons representing maximum travel extents at each threshold.

To create mutually exclusive response bands (e.g., 0–5, 5–10, 10–15 minutes), cumulative polygons were converted into discrete rings using sequential **Erase** operations. For each threshold, the smaller drive-time polygon was subtracted from the next largest cumulative polygon (e.g., 0–10 minus 0–5 to create 5–10). This process produced non-overlapping service bands suitable for parcel-level classification and statistical comparison.

Parcel centroids were spatially joined to the resulting band polygons to assign each parcel a response-time category under each staffing scenario.



Figure 8, Policy 3 Drivetime assignment model, ArcGIS Pro

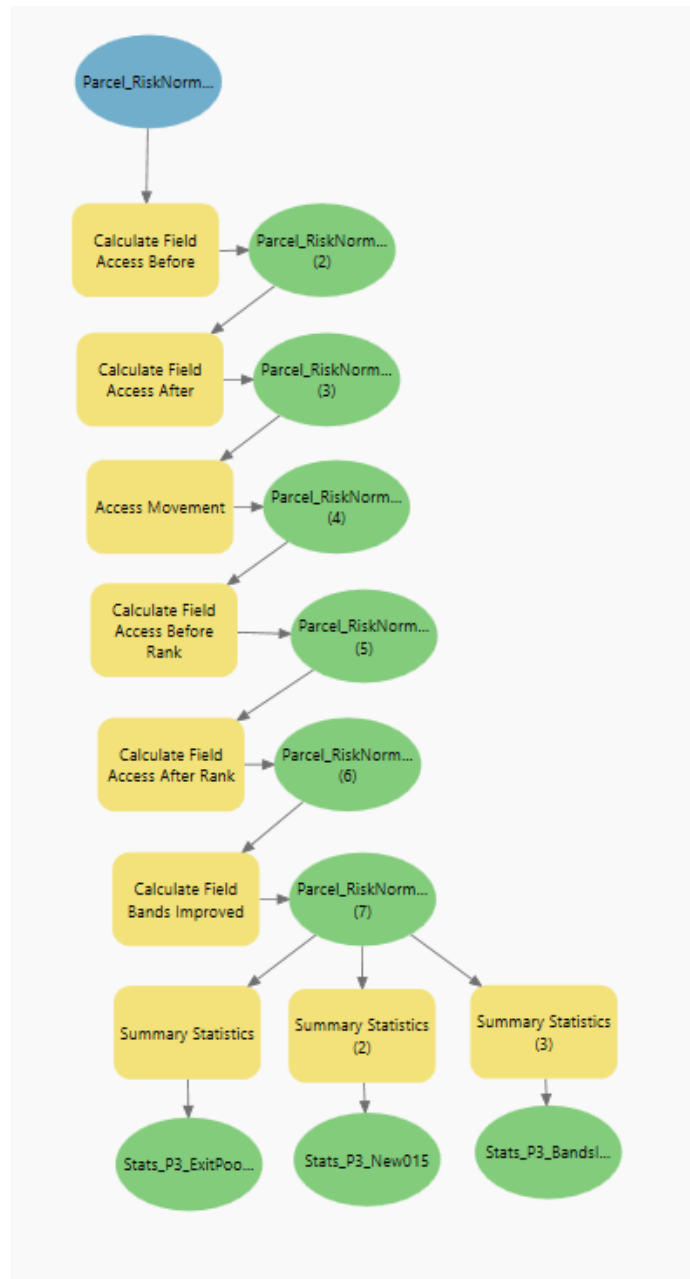


Figure 9, Policy 3 Access Improvement

Assumptions and Limitations

The Generate Drive Time Trade Areas tool relies on Esri's routing service and associated travel-speed assumptions, which may not fully reflect emergency response driving conditions or wildfire-related disruptions. The analysis assumes road network connectivity remains intact

during fire events and does not account for congestion, road closures, smoke conditions, or dispatch delay. Staffing status is modeled categorically and assumes full operational readiness of all staffed stations. Additionally, parcels within a given drive-time band are assumed to receive comparable benefit, though real-world suppression effectiveness varies based on terrain, water supply, apparatus availability, and incident complexity.

Chapter 3. Results

For policy evaluation, three baseline metrics were carried forward across all scenarios:

(1) parcels exceeding a defined wildfire hazard threshold (PRH), (2) manufactured-home parcels within high-risk or delayed-response zones, and (3) parcels outside a ≤ 15 -minute wildfire response time.

3.1 Policy 1 Results Targeted fuels work near homes (TPN hotspots)

Policy 1 concentrates fuel-treatment effort in the locations where landscape-scale mitigation is modeled to most reduce structure-relevant wildfire risk. The target cohort is defined as the top quintile of Treatment Priority near Homes (TPN) parcels that also meet exposure criteria ($TPN \geq 0.007040$; $n = 4,554$ parcels). This policy is designed to maximize countywide risk reduction per treated parcel by focusing on areas where hazardous fuels interact most directly with residential exposure pathways.

As summarized in Table 3.1, the modeled treatment effect reduces mean parcel risk to homes (PRH) from 0.01910 (before) to 0.01842 (after) across the treated set. This change

reflects the application of a 30% planning-level hazard reduction factor to parcels in the top TPN quintile (see Chapter 2 Methods). At county scale, the cumulative modeled improvement across treated parcels is 26.09 PRH “risk units” ($\Sigma\Delta\text{PRH_P1}$), representing aggregate reduction in structure-relevant wildfire exposure attributable to the targeted treatment pattern. Importantly, the treated cohort includes 608 manufactured-home parcels, indicating that a substantial portion of the modeled benefits accrue to more vulnerable housing types within the WUI.

Interpreted as a planning outcome, Policy 1 produces a broad, moderate reduction in mean PRH across a relatively large treatment footprint, and it does so by prioritizing treatment leverage (TPN) rather than targeting only the highest-risk homes. This approach is most appropriate where county-level fuel budgets are limited and decision-makers need a defensible strategy for placing treatments where they measurably reduce structure exposure (e.g., shaded fuel breaks, thinning, and prescribed fire where feasible). Results are modeled and illustrative; they assume uniform treatment effectiveness within the target set and do not represent site-specific implementation constraints, permitting, or landowner participation.

3.1a Policy 1 Metrics

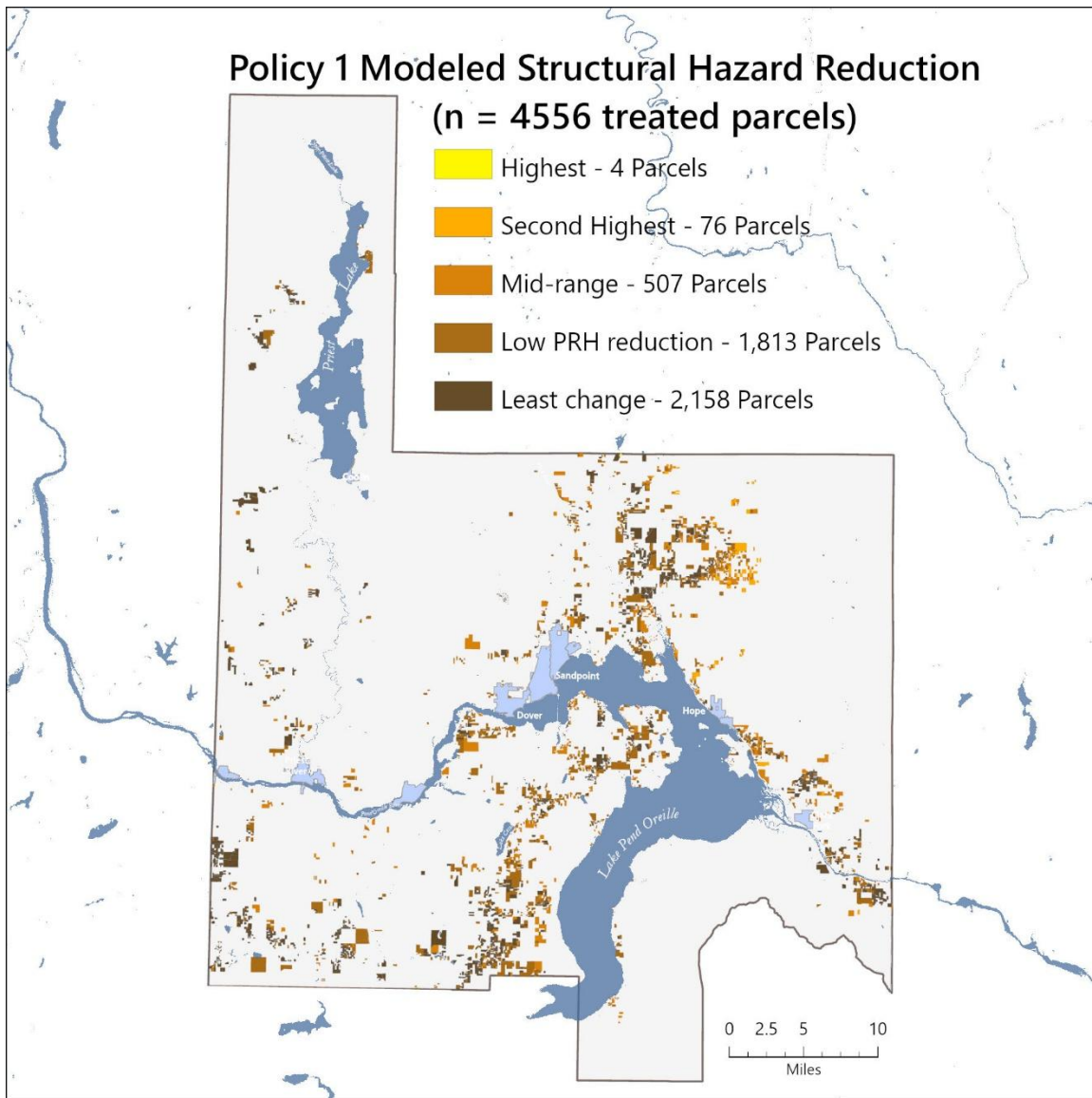


Figure 10, Policy 1 Map

Table 3.1. Policy 1 outcomes for top-quintile TPN parcels (TPN $\geq$ 0.007040).

Metric	Value
Target parcels treated	4,554
Mean PRH (Before)	0.01910
Mean PRH (After)	0.01842
Aggregate PRH reduction ($\Sigma \Delta$ PRH P1)	26.09 risk units
Manufactured-home parcels treated	608

Table 9, Policy 1 outcomes for top-quintile TPN parcels

3.2 Policy 2 - Home Hardening at Highest-Risk Homes

Treating the highest-risk homes directly ($n = 970$) yields a consistent 20% reduction in parcel hazard across the treated cohort. This reduction reflects the application of a uniform 20% planning-level hazard reduction factor to residential parcels in the top PRH quintile (see Chapter 2 Methods). Average PRH decreases from 0.01897 to 0.01518 ($\Delta \sim 0.00379$), resulting in a total modeled reduction of approximately 3.68 PRH units. Because this intervention targets only the highest-risk residential parcels, its aggregate hazard reduction is smaller than Policy 1, but benefits are concentrated on structure-specific vulnerability.

In contrast to Policy 1's more spatially concentrated gains, Policy 2 delivers incremental risk reductions across dispersed high-risk homes, making it especially relevant where landscape treatments are infeasible or constrained. This policy complements Policy 1 by reducing structure ignitability within the WUI rather than altering surrounding fuel conditions.

3.2a Metrics

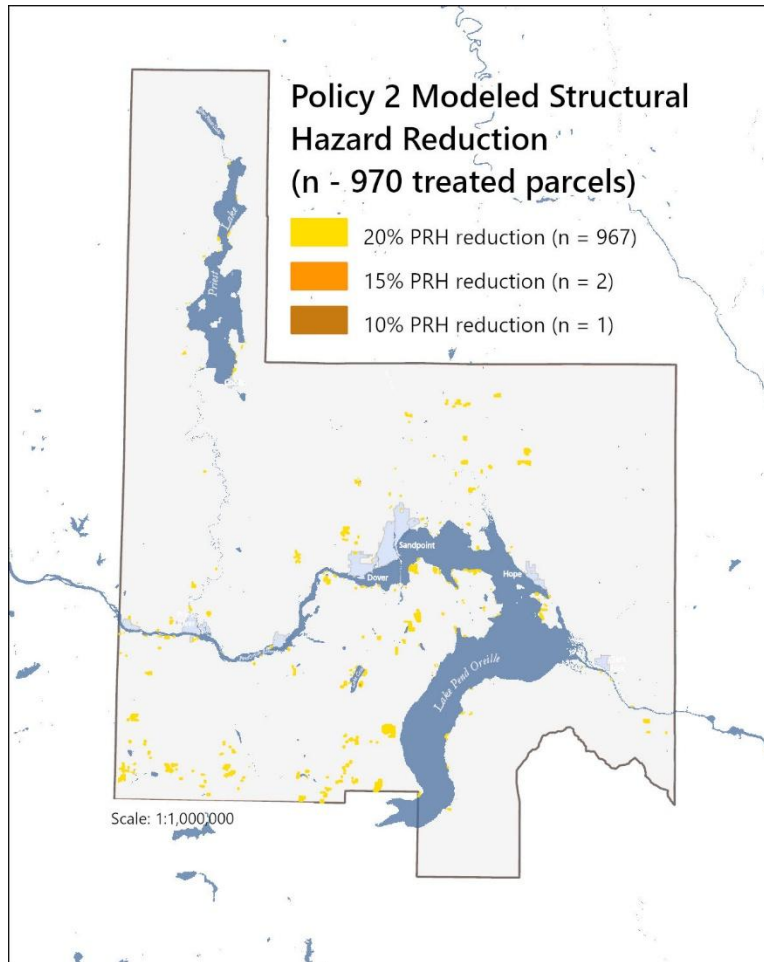


Figure 11, Policy 2 Map

Table 3.2. Policy 2 outcomes for top-quintile PRH parcels ($PRH \geq 0.013095$).

Metric	Value
Target parcels treated	970
Mean PRH (Before)	0.01897
Mean PRH (After)	0.01518
Mean ΔPRH (per parcel)	0.00379
Mean percent reduction	20%
Aggregate PRH reduction ($\Sigma \Delta PRH$ P2)	3.68 risk units

Table 10, Policy 2 outcomes for top-quintile PRH parcels

3.3 Policy 3 - Staff all stations (access improvement)

Staffing all fire stations and restoring recent closures produces a clear step-change in emergency access across Bonner County. Under current staffing, 18,272 parcels with structures fall within a 0–15 minute response window. Under a fully staffed scenario, 6,248 additional

parcels move into the ≤ 15 -minute band, increasing the total to 24,520 parcels within 15 minutes of a staffed station.

Of the parcels newly brought within 15 minutes, 1,786 are manufactured home parcels, reducing emergency isolation for housing types that are often more structurally vulnerable and evacuation-sensitive. Most improved parcels gain one or two response bands, while a smaller subset experience larger accessibility gains, reflecting meaningful reductions in response-time disparity across rural service areas.

3.3a Policy 3 Metrics

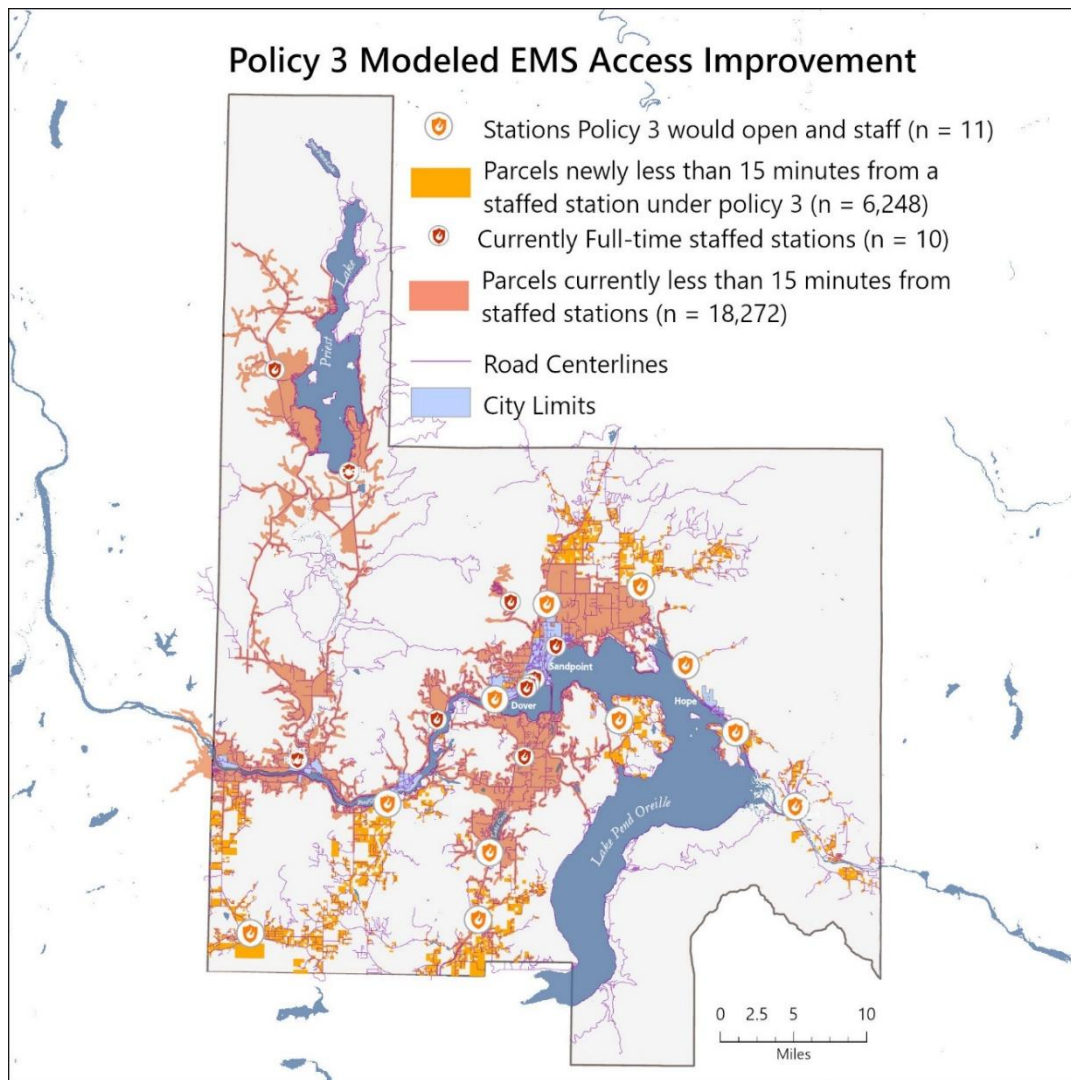


Figure 12, Policy 3 Modeled Access Improvement

Table 3.3. Policy 3 Emergency Response Improvements (Residential Parcels)

Metric	Value
Residential parcels > 15 minutes	6,625
Newly within 0–15 minutes	6,248
Parcels remaining > 15 min	377
Baseline MH < 15	545
Manufactured homes newly within 15 min	1,786
Policy 3 MH < 15	2,331
Percent reduction in >15 min parcels	94.3%
MH remaining > 15	847

Table 11, Policy 3 Emergency Response Improvements

3.4 Cost & Benefit Framing

This section provides planning-level cost benchmarks for each policy to support comparative evaluation in Chapter 4. Values are illustrative and intended for relative comparison rather than detailed budgeting. Cost-effectiveness metrics are normalized using the modeled outcomes reported in Tables 3.1–3.3, including aggregate PRH reduction and access-band improvements.

Policy 1: Targeted fuels treatments (TPN Q5 near homes)

Policy 1 assumes a 0.5-acre treatment footprint per targeted parcel, representing defensible-space and near-home fuel reduction rather than whole-parcel treatment. Treating the 4,554 highest priority near-home parcels requires an estimated \$1.8 - \$5.7M in upfront investment (\$400 - \$1,250 per parcel), with an optional 25% refresh in year 5. Normalized by the observed reduction in Parcel Risk to Homes ($\Delta PRH \sim 26.09$), this equates to approximately \$70k - \$218k per PRH point reduced across the treated set. Policy 1 delivers the largest per-dollar reduction in structure-relevant risk, particularly where wildfire hazard and exposure coincide.

Policy 2: Home hardening at highest-risk homes (PRH Q5)

Hardening the 970 highest-risk homes costs approximately \$3.4M to \$19.4M (\$3,500- \$20,000 per home), depending on retrofit. The policy applies a modeled 20% hazard reduction to each treated parcel, yielding an aggregate reduction of approximately 3.68 PRH units. This corresponds to roughly \$0.9M to \$5.3M per PRH point reduced. Because the intervention targets structure-specific vulnerability rather than landscape exposure, benefits are concentrated among the highest-risk residential parcels. Approximately 13% of treated homes are manufactured housing, suggesting that prioritizing lighter retrofit packages in lower-income or mobile-home clusters may improve equity outcomes.

Table 3.4a - Capital policies (P1 & P2): cost per unit normalized by risk reduction (ΔPRH)

Policy	Option	Unit assumption	Parcels	Initial cost (\$)	5-yr refresh (\$)	Per PRH unit cost	Notes
P1	Low	\$800/parcel	4,554	\$1,821,600	\$455,400	\$69.8k / PRH-pt	Highest leverage near homes
P1	Mid	\$1,500/parcel	4,554	3,415,500	\$853,875	\$130.9k / PRH-pt	
P1	High	\$2500/parcel	4,554	5,692,500	1,423,125	\$218.2k / PRH-pt	
P2	Lite	\$3,500/parcel	970	\$3,395,000	-	\$0.92M / PRH-pt	Benefits
P2	Standard	\$9,500/home	970	\$9,215,000	-	\$2.51M / PRH-pt	
P2	Full	\$20,000/home	970	\$19,400,000	-	\$5.28M / PRH-pt	

Table 12, Capital policies (P1 & P2): cost per unit normalized by risk reduction (Δ PRH)

Policy 3 - Station staffing (access improvement)

Policy 3 differs fundamentally from Policies 1 and 2: it represents recurring operational spending that buys time rather than directly reducing hazard. A single 2-person 24/7 rural station costs approximately \$0.50M per year, while a 3-person station costs approximately \$0.75M per year. Under the fully staffed scenario, 6,248 parcels are newly brought within a 15-minute response window beyond areas already served by existing stations. Of these newly served parcels, 924 are manufactured home parcels, highlighting the equity implications of expanded staffing. Cost-effectiveness is therefore evaluated using metrics of access, rather than hazard reduction.

Table 3.4b - Operational policy (P3): cost per access improvement

Staffing scenario (all 11 volunteer stations)	Annual program cost (\$/yr)	Cost per parcel newly ≤ 15 (n = 6,248)	Cost per MH newly ≤ 15 (n = 924)
2-person 24/7 staffing	\$5,500,000	\$880	\$5,952
3-person 24/7 staffing	\$8,250,000	\$1,321	\$8,929

Table 13, Operational policy (P3): cost per access improvement

- P1 buys landscape risk reduction near homes at \$400-\$1,250 per parcel (one-time), producing the largest aggregate PRH reduction per dollar.
- P2 buys structure survivability at \$3.5k-\$20k per home, delivering a modeled 20% hazard reduction at 970 highest-risk residences.
- P3 buys faster initial attack and evacuation at \$880-\$1,321 per parcel newly within fifteen minutes per year.

Comparative Summary – All Policies

Metric	Policy 1 – TPN Fuels	Policy 2 – Home Hardening	Policy 3 – Station Staffing
Baseline reference			
Target population	4,554 parcels	970 highest-risk homes	30,957 residential parcels evaluated
Primary benefit type	Landscape risk reduction	Structure ignitability reduction	Emergency response time reduction
Mean PRH (Before)	0.01910	0.01897	–
Mean PRH (After)	0.01842	0.01518	–
Aggregate PRH reduction	26.09 risk units	3.68 risk units	–
Parcels newly ≤ 15 min	–	–	6,248
Manufactured homes benefiting	608 treated	~13% of treated homes	924 newly ≤ 15 min
Capital / annual cost	\$1.8–5.7M (one-time + refresh)	\$3.4–19.4M (one-time)	\$5.5–8.25M per year
Cost efficiency metric	\$70k–218k per PRH unit	\$0.9M–5.3M per PRH unit	\$880–1,321 per parcel ≤ 15 (annual)

Table 14, Comparative Summary – All Policies

Chapter 4. Recommendations

4.1 Sequenced Investment Strategy

Based on the comparative results in Chapter 3 (Tables 3.1–3.4), a phased wildfire resilience strategy is recommended:

Phase 1 (Immediate Operational Stabilization):

Restore and maintain 24/7 staffing (Policy 3). As shown in Table 3.3, staffing brings 6,248

additional parcels — including 924 manufactured homes — within a 15-minute response window. These gains are immediately realized upon implementation and are not contingent on landowner participation or capital retrofits. By improving time-to-response and crew availability, staffing increases initial attack probability and reduces emergency isolation in rural portions of the county.

Phase 2 (High-Leverage Hazard Reduction):

Implement TPN Q5 treatments (Policy 1). Table 3.1 shows an aggregate PRH reduction of 26.09 across 4,554 parcels, representing the largest modeled structure-relevant risk reduction per capital dollar.

Phase 3 (Targeted Structure Hardening):

Home hardening at the highest-risk homes (Policy 2) delivers consistent hazard reduction within the most vulnerable residential parcels (see Table 3.2). Although less cost-efficient on a per-unit basis than Policy 1, it directly reduces structure ignitability and complements landscape treatments, particularly where fuel work is constrained.

4.2 Indicators and Monitoring Framework

- % of high-risk parcels treated
- Change in parcels >15 min response
- Treatment equity ratio (MH vs non-MH)
- Annual PRH reduction trend
- Fire district staffing growth rate

4.3 Equity & maladaptation guardrails

- Avoid “just green enough”/aesthetic fuels work that displaces risk or privileges affluent edges; combine treatments with access improvements, home hardening, and support for low-income households (Hersher & Benincase, 2019).
- Monitor for uneven benefits (e.g., who actually moved into 0 -15; who received parcel-scale mitigation).
- Engage local Volunteer Fire Departments and Homeowner Associations to ensure implementation doesn’t shift costs onto the least resourced residents.

Conclusion

This report demonstrates that wildfire risk in Bonner County is not a singular problem but a coupled system of landscape hazard, structural exposure, and emergency access, each responding to different policy levers. By distinguishing where wildfire is likely, where it threatens homes, and who is most affected, the analysis moves beyond generic mitigation toward decision-relevant prioritization grounded in parcel-scale metrics.

The results show that targeted near-home fuel treatments (Policy 1) generate the strongest aggregate reduction in structure-relevant risk among capital strategies. Staffing restoration and 24/7 minimums (Policy 3) produce immediate, system-wide improvements in emergency access, substantially expanding rapid-response coverage across rural areas and reducing isolation for manufactured housing. Home hardening at the highest-risk residences (Policy 2) delivers consistent parcel-level hazard reduction, directly lowering structure ignitability and complementing landscape treatments where fuels work is constrained.

Taken together, these findings support a sequenced resilience strategy: first buying time through staffing, then reducing leverage points of exposure near homes, and reinforcing

survivability at the most vulnerable structures, all while applying an explicit equity lens to ensure benefits accrue to manufactured housing and lower-income WUI communities.

Rather than offering a single solution, this framework provides Bonner County with a transparent and scalable approach to wildfire resilience. It aligns investments with measurable outcomes — minutes gained, risk reduced, and vulnerable parcels protected — and avoids the maladaptive tendency to privilege visible treatments over demonstrable effectiveness

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